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SYSTEM ON CHIP, METHOD FOR OPERATING THEREOF AND VIDEO SYSTEM

Abstract

A system on chip includes: a codec encoding image frames; and a codec controller sending the image frames to the codec and allocating a buffer for encoding the image frames, The codec controller sends a first image frame to the codec. The codec controller allocates the first buffer to a memory and sends the first image frame to the codec. The codec encodes the first image frame and sends the encoded first image frame to the codec controller. The codec controller allocates the first buffer to the memory and sends a second image frame to the codec. The codec requests the codec controller to allocate a second buffer of a second size. The codec controller allocates the second buffer to the memory and sends the second image frame to the codec, and the codec encodes the second image frame and sends the encoded second image frame to the codec controller.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority under 35 USC § 119 to Korean Patent Application No. 10-2024-0030848 filed on Mar. 4, 2024 in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] The present inventive concept relates to a system on a chip (SoC), a method for operating thereof, and a video system.

DISCUSSION OF THE RELATED ART

[0003] There are a variety of electronic devices that use or display still and video images, and the amount of image data used in the electronic devices has significantly increased as technology has developed and the transmission of information has increased. These significant amounts of video content and image data are broadcast to various electronic devices such as mobile devices. The development in video encoding/decoding technology enables sending of more image data through channels with limited bandwidth, and enables storage of video data with high resolution within limited storage capacity. More efficient video coding technologies may be desirable with the increased use of video content.

SUMMARY

[0004] According to exemplary embodiments of the present inventive concept, a system on chip (SoC) includes: a codec configured to encode a plurality of image frames; and a codec controller configured to send the plurality of image frames to the codec and to allocate a buffer for encoding the plurality of image frames in a memory, wherein the codec controller sends a first image frame with a first resolution to the codec without allocating the buffer, the codec requests the codec controller to allocate a first buffer of a first size that corresponds to the first resolution, in response to receiving the first image frame, the codec controller allocates the first buffer to the memory and sends the first image frame to the codec, the codec encodes the first image frame by using the first buffer and then sends the encoded first image frame to the codec controller, the codec controller allocates the first buffer to the memory and sends a second image frame with a second resolution that is smaller than the first resolution to the codec, the codec requests the codec controller to allocate a second buffer of a second size that is smaller than the first size, in response to receiving the second image frame, the codec controller allocates the second buffer to the memory and sends the second image frame to the codec, and the codec encodes the second image frame by using the second buffer and then sends the encoded second image frame to the codec controller.

[0005] According to exemplary embodiments of the present inventive concept, a method for operating a system on chip (SoC) includes a codec that is configured to sequentially encode a plurality of image frames, and a codec controller that is configured to send the plurality of image frames to the codec and allocate a buffer for encoding the plurality of image frames in a memory, the method includes: sending a first image frame with a first resolution to the codec without allocating the buffer, through the codec controller; requesting, through the codec, the codec controller to allocate a first buffer of a first size corresponding to the first resolution, in response to receiving the first image frame; allocating the first buffer to the memory and sending the first image frame to the codec, through the codec controller; encoding the first image frame by using the first buffer and then sending the encoded first image frame to the codec controller, through the codec; allocating the first buffer to the memory and sending a second image frame with a second

resolution that is smaller than the first resolution to the codec, through the codec controller; requesting the codec controller to allocate a second buffer of a second size that is smaller than the first size, in response to receiving the second image frame; allocating the second buffer to the memory and sending the second image frame to the codec; and encoding the second image frame by using the second buffer and then sending the encoded second image frame to the codec controller.

[0006] According to exemplary embodiments of the present inventive concept, a video system includes: a codec configured to encode a plurality of image frames; a codec controller configured to send the plurality of image frames to the codec and to allocate a buffer for encoding the plurality of image frames in a memory; a camera configured to send the plurality of image frames to the codec controller; and a display configured to display the plurality of image frames, wherein the codec controller sends a first image frame with a first resolution to the codec without allocating the buffer, the codec requests the codec controller to allocate a first buffer of a first size that corresponds to the first resolution, in response to receiving the first image frame, the codec controller allocates the first buffer to the memory and sends the first image frame to the codec, the codec encodes the first image frame by using the first buffer and then sends the encoded first image frame to the codec controller, the codec controller allocates the first buffer to the memory and sends a second image frame with a second resolution that is smaller than the first resolution to the codec, the codec requests the codec controller to allocate a second buffer of a second size that is smaller than the first size, in response to receiving the second image frame, the codec controller allocates the second buffer to the memory and sends the second image frame to the codec, and the codec encodes the second image frame by using the second buffer and then sends the encoded second image frame to the codec controller.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The above and other aspects and features of the present inventive concept will become more apparent by describing in detail exemplary embodiments thereof with reference to the attached drawings, in which:

[0008] FIG. 1 is a block diagram for describing a video system according to exemplary embodiments of the present inventive concept.

[0009] FIG. 2 is a block diagram for describing some components of the system on chip illustrated in FIG. 1.

[0010] FIG. 3 is a block diagram for describing the codec illustrated in FIG. 2.

[0011] FIG. 4 is a block diagram for describing an encoder illustrated in FIG. 3.

[0012] FIG. 5 is a block diagram for describing a decoder illustrated in FIG. 3.

[0013] FIG. 6 is a block diagram for describing the DPB illustrated in FIG. 1.

[0014] FIG. 7 is a block diagram illustrating only some components of FIG. 1 to describe an operation of the video system illustrated in FIG. 1.

[0015] FIG. 8 is a block diagram illustrating only some components of FIG. 1 to describe an operation of the video system illustrated in FIG. 1.

[0016] FIG. 9 is a flowchart describing an operation of a video system according to exemplary embodiments of the present inventive concept.

[0017] FIG. 10 is a table for describing the effects of a video system according to exemplary embodiments of the present inventive concept.

[0018] FIG. 11 is a flowchart for describing an operation of a video system according to exemplary embodiments of the present inventive concept.

[0019] FIG. 12 is a table for describing the effects of a video system according to exemplary

embodiments of the present inventive concept.

[0020] FIG. **13** is a flowchart for describing an operation of a video system according to exemplary embodiments of the present inventive concept.

[0021] FIG. **14** is a flowchart for describing an operation of a video system according to exemplary embodiments of the present inventive concept.

[0022] FIG. **15** is an exemplary diagram for describing an electronic device including a video system according to exemplary embodiments of the present inventive concept.

[0023] FIG. **16** is a diagram describing a network system between electronic devices including a video system according to exemplary embodiments of the present inventive concept.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0024] Hereinafter, exemplary embodiments of the present inventive concept will be described in detail with reference to the accompanying drawings. The same reference numerals are used for the same components in the drawings and specification, and duplicate descriptions thereof may be omitted or briefly discussed.

[0025] FIG. **1** is a block diagram for describing a video system according to exemplary embodiments of the present inventive concept.

[0026] Referring to FIG. **1**, a video system **1000** according to exemplary embodiments of the present inventive concept may include a system on chip (SoC) **10** for encoding or decoding video (hereinafter referred to as system on chip for convenience of explanation), a camera **300**, a memory **500**, and a display **700**.

[0027] In the video system **1000** according to exemplary embodiments of the present inventive concept, the memory **500** is disposed outside the system on chip **10**, but the present inventive concept is not limited thereto, and for example, the memory **500** may be implemented inside the system on chip **10**.

[0028] The system on chip **10** may generally control an operation of the video system **1000**. For example, the system-on-chip **1000** may be an integrated circuit (IC), a motherboard, an application processor (AP), or a mobile AP.

[0029] The system on chip **10** may process image data that is output from the camera **300**, such as image frames, to generate a stream, and may display the processed stream through the display **700** and/or store the processed stream in the memory **500**.

[0030] The system on chip **10** may include a codec controller **100**, a codec **200**, a memory controller **400**, and a display controller **600**.

[0031] The codec controller **100**, the codec **200**, the memory controller **400**, and the display controller **600** may exchange data with each other through a bus **800**. For example, the bus **800** may be implemented as a peripheral component interconnect (PCI) bus, a PCI express bus, an advanced microcontroller bus architecture (AMBA), an advanced high performance bus (AHB), an advanced peripheral bus (APB), or an advanced extensible interface (AXI) bus, or a combination thereof.

[0032] The codec **200** may encode frames that are obtained from the camera **300** or frames that are stored in the memory **500**, or may decode streams that are stored in the memory **500**. For example, the camera **300** may be implemented as a complementary metal oxide semiconductor (CMOS) image sensor.

[0033] The system on chip **10** may further include, for example, a central processing unit (CPU), a neural processing unit (NPU), a graphics processing unit (GPU), a digital signal processing processor (DSP), etc., but the present inventive concept is not limited thereto.

[0034] The display controller **600** may display the image frame that is output from the codec controller **100** on the display **700**. For example, the display **700** may be implemented as a monitor, a TV monitor, a projection device, a thin film transistor-liquid crystal display (TFT-LCD), a light emitting diode (LED) display, an organic LED (OLED) display, an active-matrix OLED (AMOLED) display, or a flexible display, but the present inventive concept is not limited thereto.

[0035] For example, the display controller **600** may send the image data to the display **700** through a display serial interface (DSI).

[0036] The memory controller **400** may read the data that is stored in the memory **500** based on a command output from the codec controller **100** and may send the read data to the codec controller **100**. In addition, the memory controller **400** may write data that is output from the codec controller **100** to the memory **500** based on the command that is output from the codec controller **100**.

[0037] According to exemplary embodiments of the present inventive concept, the memory **500** may be implemented as a volatile memory or a non-volatile memory. The volatile memory may be implemented as a random access memory (RAM), a static RAM (SRAM), a dynamic RAM (DRAM), a synchronous DRAM (SDRAM), a thyristor RAM (T-RAM), a zero capacitor RAM (Z-RAM), or a twin transistor RAM (TTRAM), but the present inventive concept is not limited thereto.

[0038] For example, the non-volatile memory may be implemented as an electrically erasable programmable read-only memory (EEPROM), a flash memory, a magnetic RAM (MRAM), a spin-transfer torque MRAM, a ferroelectric RAM (FeRAM), a phase change RAM (PRAM), or a resistive RAM (RRAM). In addition, the non-volatile memory may be implemented as, for example, a multimedia card (MMC), an embedded MMC (eMMC), a universal flash storage (UFS), a solid state drive (SSD), a USB flash drive, or a hard disk drive (HDD), but the present inventive concept is not limited thereto.

[0039] The memory **500** may include a decoded picture buffer (DPB) **550**. When encoding or decoding is performed in the system on chip **10**, the image frames may be stored in the DPB **550** in the form of reference frames. The frames that are stored in the DPB **550** may be used to encode or decode video. The DPB **550** may be implemented as, for example, a random access memory (RAM), a static RAM (SRAM), a dynamic RAM (DRAM), a synchronous DRAM (SDRAM), a thyristor RAM (T-RAM), a zero capacitor RAM (Z-RAM), or a twin transistor (TTRAM), but the present inventive concept is not limited thereto.

[0040] FIG. **2** is a block diagram for describing some components of the system on chip illustrated in FIG. **1**.

[0041] Referring to FIG. **2**, the system on chip **10** may include a codec controller **100** and a codec **200**. The codec controller **100** may receive input data IN from other components (e.g., the camera **300**, the memory **500**, etc.) that are within the video system **1000**. However, the codec controller **100** might not directly encode or decode the input data IN. Therefore, the codec controller **100** may send the input data IN to the codec **200**.

[0042] The codec **200** may directly encode or decode the input data IN received from the codec controller **100**. The input data IN may be an image frame with arbitrary or predetermined resolution. For example, the resolution may be 2K, 4K, 5K, or 8K, but the exemplary embodiment of the present inventive concept is not limited thereto. However, the codec **200** might not directly allocate the DPB **550**. Therefore, when a buffer of a size that is different from the resolution of the input data IN is allocated, the codec **200** may request the codec controller **100** to allocate a buffer of a new size.

[0043] The codec controller **100** may allocate the DPB **550** having the same size as the resolution of the input data IN and send the input data IN to the codec **200**. Output data OUT encoded or decoded by the codec **200** may be sent to the codec controller **100**, and the codec controller **100** may send the output data OUT to other components (e.g., memory **500**, display **700**, etc.) that are within the video system **1000**.

[0044] For example, the input data IN may be an image frame sent from the camera **300**. The codec controller **100** may send the image frame to the codec **200**. When allocation of the DPB **550** with a new size is required, the codec **200** may request such allocation from the codec controller **100**. For example, the codec **200** may request the codec controller **100** to allocate a buffer of the same size as the resolution of the image frame. The codec controller **100** may allocate the buffer of the

requested size in the DPB **550** and send the image frame to the codec **200**. The codec **200** may send an encoded image frame that is obtained by encoding the image frame to the codec controller **100**. The codec controller **100** may store the encoded image frame in the memory **500**. However, the exemplary embodiment of the present inventive concept is not limited thereto.

[0045] For example, the input data IN may be the encoded image frame sent from the memory **500**. The codec controller **100** may send the encoded image frame to the codec **200**. When allocation of the DPB **550** with a new size is required, the codec **200** may request such allocation from the codec controller **100**. For example, the codec **200** may request the codec controller **100** to allocate a buffer of the same size as the resolution of the image frame. The codec controller **100** may allocate the buffer of the requested size in the DPB **550** and send the encoded image frame to the codec **200**. The codec **200** may send an image frame obtained by decoding the encoded image frame to the codec controller **100**. The codec controller **100** may display the image frame on the display **700**. However, the exemplary embodiment of the present inventive concept is not limited thereto.

[0046] FIG. **3** is a block diagram for describing the codec illustrated in FIG. **2**. FIG. **4** is a block diagram for describing an encoder illustrated in FIG. **3**. FIG. **5** is a block diagram for describing a decoder illustrated in FIG. **3**.

[0047] Referring to FIGS. **3** to **5**, the codec **200** may include an encoder **210** and a decoder **230**. The encoder **210** may encode the image frame that is sent from the codec controller **100** and may send the encoded image frame to the codec controller **100**. The decoder **230** may decode the encoded image frame that is sent from the codec controller **100**, and may send the decoded image frame to the codec controller **100**.

[0048] The encoder **210** may include a bitstream generation unit (CR_BS) (e.g., a bitstream generation circuit) **2110** and an encoding unit (e.g., an encoding circuit) **2130**. The bitstream generation unit **2110** and the encoding unit **2130** may include at least one processor. In addition, the bitstream generation unit **2110** and the encoding unit **2130** may include a memory that stores instructions that are to be performed by at least one processor.

[0049] The encoding unit **2130** may receive an image frame and encode the image frame to generate an encoded image frame. The bitstream generation unit **2110** may generate a bitstream based on the encoded image frame. The bitstream generated by the bitstream generation unit **2110** may be stored in the memory **500**.

[0050] The decoder **230** may include a bitstream acquisition unit (AC_BS) **2310** and a decoding unit (e.g., a decoding circuit) **2330**. For example, the bitstream acquisition unit **2310** and the decoding unit **2330** may include at least one processor. In addition, the bitstream acquisition unit **2310** and the decoding unit **2330** may include a memory that stores instructions that are to be performed by at least one processor.

[0051] The bitstream acquisition unit **2310** may receive a bitstream. The bitstream includes the encoded image frame that is obtained by encoding the image frame by the above-described encoding unit **2130**. The bitstream acquisition unit **2310** may receive a bitstream from the memory **500**. The decoding unit **2330** may restore the encoded image frame based on information that is obtained from the received bitstream.

[0052] FIG. **6** is a block diagram for describing the DPB illustrated in FIG. **1**.

[0053] Referring to FIG. **6**, when encoding or decoding is performed in the system on chip **10**, the DPB may store the image frames in the form of reference frames. The image frames stored in the DPB **550** may be used to encode or decode video.

[0054] The codec controller **100** may allocate a buffer with an arbitrary or predetermined size within the DPB **550**. For example, the codec controller **100** may allocate a buffer of a size corresponding to a first area **5501**. The size of the first area **5501** may be, for example, 64 k*64 k (hereinafter, referred to as 8K). The codec controller **100** may allocate a buffer of a size corresponding to a second area **5502**. The size of the second area **5502** may be, for example, 2048 k*1080 k (hereinafter, referred to as 2K). The codec controller **100** may allocate a buffer of a size

corresponding to a third area **5503**. The size of the third area **5503** may be, for example, $3840\text{ k} \times 2160\text{ k}$ (hereinafter, referred to as 4K). The codec controller **100** may allocate a buffer of a size corresponding to a fourth area **5504**. The size of the fourth area **5504** may be, for example, $5120\text{ k} \times 2160\text{ k}$ (hereinafter, referred to as 5K).

[0055] As described above with reference to FIG. 2, the codec controller **100** may allocate the buffer with an arbitrary or predetermined size within the DPB **550**.

[0056] Buffers corresponding to the first to fourth areas **5501** to **5504** may store image frames with a resolution smaller than or equal to the size of each buffer. When the resolution of the image frame is greater than the size of the buffer, a page fault error may occur.

[0057] In the drawing, only the first to fourth areas **5501** to **5504** having different sizes are illustrated, but the exemplary embodiment of the present inventive concept is not limited thereto. For example, the DPB **550** may also include a buffer corresponding to an area different from the first to fourth areas **5501** to **5504**.

[0058] In addition, in the drawing, the first to fourth areas **5501** to **5504** are illustrated as overlapping, but the exemplary embodiment of the present inventive concept is not limited thereto. For example, a plurality of buffers in the DPB **550** may also be allocated without overlapping each other.

[0059] FIG. 7 is a block diagram illustrating only some components of FIG. 1 to describe an operation of the video system illustrated in FIG. 1.

[0060] Referring to FIG. 7, when performing an encoding operation, the operations of the codec controller **100**, the codec **200**, and the DPB **550** are as follows.

[0061] The image frame FR may be sent to the codec controller **100**. For example, the image frame FR may be sent from the memory **500** or from the camera **300**. The image frame FR may be sent to the DPB **550** and may be used in the form of a reference frame when the codec **200** encodes the image frame FR.

[0062] In this case, when the image frame FR is sent to the codec **200** without allocating the buffer within the DPB **550**, or when the size of the buffer in the DPB **550** is smaller than the resolution of the image frame FR, the codec **200** may request the codec controller **100** to allocate a buffer of a new size.

[0063] In addition, when the size of the buffer in the DPB **550** is greater than or equal to the resolution of the image frame FR, the image frame FR may be encoded by the codec **200**. An encoded image frame EN_FR, which is formed by being encoded by the codec **200**, may be sent to the codec controller **100**. The codec controller **100** may send the encoded image frame EN_FR to another component (e.g., memory **500**) within the video system **1000**.

[0064] FIG. 8 is a block diagram illustrating only some components of FIG. 1 to describe an operation of the video system illustrated in FIG. 1. Since FIG. 8 is similar to FIG. 7, differences will be mainly described and redundant descriptions may be omitted. To the extent that the description of various elements are omitted, it may be assumed that these elements are at least similar to corresponding elements that have already been described.

[0065] Referring to FIG. 8, when performing a decoding operation, the operations of the codec controller **100**, the codec **200**, and the DPB **550** are as follows.

[0066] The encoded image frame EN_FR may be sent to the codec controller **100**. For example, the encoded image frame EN_FR may be sent from the memory **500**. The encoded image frame EN_FR decoded by the codec **200** may be sent to the DPB **550** and used in the form of a reference frame when the codec **200** decodes the encoded image frame EN_FR.

[0067] In this case, when the encoded image frame EN_FR is sent to the codec **200** without allocating the buffer within the DPB **550**, or the size of the buffer in the DPB **550** is smaller than the resolution of the encoded image frame EN_FR, the codec **200** may request the codec controller **100** to allocate a buffer of a new size.

[0068] In addition, when the size of the buffer in the DPB **550** is greater than or equal to the

resolution of the encoded image frame EN_FR, the encoded image frame EN_FR may be decoded by the codec **200**. An image frame FR that is formed by being decoded by the codec **200** may be sent to the codec controller **100**. The codec controller **100** may send the image frame FR to another component (e.g., display) within the video system **1000**.

[0069] FIG. **9** is a flowchart for describing an operation of a video system according to exemplary embodiments of the present inventive concept. The specific values of the buffer size and frame resolution described in FIG. **9** are examples for convenience of explanation, and the exemplary embodiment of the present inventive concept is not limited thereto.

[0070] Referring to FIGS. **7** and **9**, an operation of a video system according to exemplary embodiments of the present inventive concept may be as follows.

[0071] The codec controller **100** may send a first image frame FR1 with a first resolution (e.g., may be 64 k*64 k, hereinafter, referred to as 8K) to the codec **200** without allocating the DPB **550** (S901).

[0072] The codec **200** may request a buffer of a first size (8K) corresponding to the first resolution (8K) from the codec controller **100** (S902).

[0073] The codec controller **100** may allocate the buffer of the first size (8K) in the DPB **550** and send the first image frame FR1 with the first resolution (8K) to the codec **200** (S903).

[0074] The codec **200** may send a first encoded image frame EN_FR1 obtained by encoding the first image frame FR1 to the codec controller **100** (S904).

[0075] The codec controller **100** may allocate the buffer of the first size (8K) in the DPB **550** and send a second image frame FR2 with a second resolution (e.g., may be 2048*1080, hereinafter, referred to as 2K) smaller than the first resolution (8K) to the codec **200** (S905).

[0076] The codec **200** may request a buffer of a second size (2K) corresponding to the second resolution (2K) from the codec controller **100** (S906).

[0077] The codec controller **100** may allocate the buffer of the second size (2K) in the DPB **550** and send the second image frame FR2 with the second resolution (2K) to the codec **200** (S907).

[0078] The codec **200** may send a second encoded image frame EN_FR2 that is obtained by encoding the second image frame FR2 to the codec controller **100** (S908).

[0079] The codec controller **100** may allocate the buffer of the second size (2K) in the DPB **550** and send a third image frame FR3 with a third resolution (e.g., may be 3048*2160, hereinafter, referred to as 4K) that is greater than the second resolution (2K) to the codec **200** (S909).

[0080] The codec **200** may request a buffer of a third size (4K) corresponding to the third resolution (4K) from the codec controller **100** (S910).

[0081] The codec controller **100** may allocate the buffer of the third size (4K) in the DPB **550** and send the third image frame FR3 with the third resolution (4K) to the codec **200** (S911).

[0082] The codec **200** may send a third encoded image frame EN_FR3, which is obtained by encoding the third image frame FR3, to the codec controller **100** (S912).

[0083] For example, as interaction between the codec controller **100** and the codec **200** is performed every time each image frame is sent, a buffer with a size that corresponds to the resolution of each image frame may be allocated within the DPB **550**. Through this, a wasted memory space may be minimized.

[0084] FIG. **10** is a table describing the effects of a video system according to exemplary embodiments of the present inventive concept.

[0085] Referring to FIG. **10**, each figure in the table assumes that the first image frame is 8K and that 9 image frames of which the resolution changes to 2K and 4K for each image frame are encoded or decoded. In the following, the description is based on encoding, but the same may also be applied to decoding.

[0086] For example, when a buffer of a size that corresponds to a maximum image frame size in the stream is allocated (Max), a buffer of a size of 65536*65536 (8K) is used per image frame. Therefore, when encoding 9 image frames, a buffer of a total size of 38,654,705,664 is used. Two

interactions occur between the codec controller **100** and the codec **200** for each image frame. For example, the interactions include sending an image frame from the codec controller **100** to the codec **200** and sending an encoded image frame from the codec **200** to the codec controller **100**. Therefore, when encoding 9 image frames, a total of 18 interactions occur.

[0087] In addition, according to an exemplary embodiment of the present inventive concept described with reference to FIGS. 7 and 9 (Method 1), one 8K buffer, four 2K buffers, and four 4K buffers are used. Therefore, when encoding 9 image frames, a buffer of a total size of 75,202,560 is used. As described with reference to FIG. 9, four interactions occur for each image frame between the codec controller **100** and the codec **200** (e.g., S901 to S904 in FIG. 9). Therefore, when encoding 9 image frames, a total of 36 interactions occur.

[0088] In this way, waste of the memory may be effectively prevented through the interaction between the codec controller **100** and the codec **200**.

[0089] FIG. 11 is a flowchart describing an operation of a video system according to exemplary embodiments of the present inventive concept. The specific values of the buffer size and frame resolution described in FIG. 11 are examples for convenience of explanation, and the exemplary embodiment of the present inventive concept is not limited thereto.

[0090] Referring to FIGS. 7 and 11, an operation of a video system according to exemplary embodiments of the present inventive concept may be as follows.

[0091] The codec controller **100** may send a fourth image frame FR4 with a fourth resolution (e.g., may be 64 k*64 k, hereinafter, referred to as 8K) to the codec **200** without allocating the DPB **550** (S1101).

[0092] The codec **200** may request a buffer of a fourth size (8K) corresponding to the fourth resolution (8K) from the codec controller **100** (S1102).

[0093] The codec controller **100** may allocate the buffer of the fourth size (8K) in the DPB **550** and send the fourth image frame FR4 with the fourth resolution (8K) to the codec **200** (S1103).

[0094] The codec **200** may send a fourth encoded image frame EN_FR4, which is obtained by encoding the fourth image frame FR4, to the codec controller **100** (S1104).

[0095] The codec controller **100** may allocate the buffer of the fourth size (8K) in the DPB **550** and send a fifth image frame FR5 with a fifth resolution (e.g., may be 2048 k*1080 k, hereinafter, referred to as 2K) that is smaller than the fourth resolution (8K) to the codec **200** (S1105).

[0096] The codec **200** may request a buffer of a sixth size (4K) corresponding to a sixth resolution (e.g., may be 3840*2160, hereinafter, referred to as 4K) greater than the fifth resolution (2K) from the codec controller **100** (S1106). For example, if it is possible to allocate a buffer with a size smaller than the size (e.g., a) of the allocated buffer and a size (e.g., b) greater than the resolution of the sent image frame, the codec may request a buffer of a size corresponding to the range from b to a from the codec controller. For example, a buffer of the sixth size (4K), which is smaller than the buffer of the fourth size (8K) and greater than the buffer of the fifth size (2K), may be requested as fake resolution. For example, the fake resolution refers to a buffer that the codec **200** requests from the codec controller **100**, considering a trade-off between the memory size and memory performance. In addition, a size of the fake resolution may be greater than the resolution of the image frame that the codec **200** receives from the codec controller **100**.

[0097] By using the fake resolution, the number of interactions between the codec controller **100** and the codec **200** may be reduced while efficiently using the memory space. Through this, the memory space and performance may be balanced.

[0098] The codec controller **100** may allocate the buffer of the sixth size (4K) in the DPB **550** and send the fifth image frame FR5 with the fifth resolution (2K) to the codec **200** (S1107).

[0099] The codec **200** may send a fifth encoded image frame EN_FR5 that is obtained by encoding the fifth image frame FR5 to the codec controller **100** (S1108).

[0100] The codec controller **100** may allocate the buffer of the sixth size (4K) in the DPB **550** and send a sixth image frame FR6 with a sixth resolution (e.g., may be 3048*2160, hereinafter, referred

to as 4K) to the codec **200** (S1109).

[0101] The codec **200** may send a sixth encoded image frame EN_FR6 that is obtained by encoding the sixth image frame FR6 to the codec controller **100** (S1110).

[0102] As described in S1106 to S1110, by using the fake resolution, the number of interactions between the codec controller **100** and the codec **200** may be reduced while efficiently using the memory space. Through this, the memory space and performance may be balanced.

[0103] FIG. 12 is a table for describing the effects of a video system according to exemplary embodiments of the present inventive concept.

[0104] Referring to FIG. 12, each figure in the table assumes that the first image frame is 8K and that 9 image frames of which the resolution changes to 2K and 4K for each image frame are encoded or decoded. In the following, the description is based on encoding, but the same may also be applied to decoding.

[0105] For example, when a buffer of a size corresponding to a maximum image frame size in the stream is allocated (Max), a buffer of a size of 65536*65536 (8K) is used per image frame.

Therefore, when encoding 9 image frames, a buffer of a total size of 38,654,705,664 is used. Two interactions may occur between the codec controller **100** and the codec **200** for each image frame. For example, the interactions include sending an image frame from the codec controller **100** to the codec **200** and sending an encoded image frame from the codec **200** to the codec controller **100**. Therefore, when encoding 9 image frames, a total of 18 interactions occur.

[0106] In addition, according to an exemplary embodiment of the present inventive concept described with reference to FIGS. 7 and 11 (Method 2), one 8K buffer and eight 4K buffers are used. Therefore, when encoding 9 image frames, a buffer of a total size of 99,532,800 is used. As described with reference to FIG. 11, four interactions each occur between the codec controller **100** and the codec **200** in the fourth image frame FR4 and the fifth image frame FR5 (e.g., S1101 to S1104 and S1105 to S1108 in FIG. 11). In addition, two interactions each occur in the remaining image frames. Therefore, when encoding 9 image frames, a total of 22 interactions occur.

[0107] In this way, by using the fake resolution, the number of interactions between the codec controller **100** and the codec **200** may be reduced while efficiently using the memory space. Through this, the memory space and performance may be balanced.

[0108] FIG. 13 is a flowchart for describing an operation of a video system according to exemplary embodiments of the present inventive concept. The specific values of the buffer size and frame resolution described in FIG. 13 are examples for convenience of explanation, and the exemplary embodiment of the present inventive concept is not limited thereto. A scenario in FIG. 13 corresponds to a further expanded exemplary embodiment of the scenario in FIG. 11, and S1301 to S1310 in FIG. 13 correspond to steps corresponding to S1101 to S1110 in FIG. 11, respectively. Therefore, in the following, the description will start from S1311.

[0109] Referring to FIGS. 7, 11, and 13, an operation of a video system according to exemplary embodiments of the present inventive concept may be as follows.

[0110] The codec controller **100** may allocate the buffer of the sixth size (4K) in the DPB **550** and may send a seventh image frame FR7 with a seventh resolution (e.g., may be 3048*2160, hereinafter, referred to as 4K) to the codec **200** (S1311). For example, the image frames with the same resolution as each other may be continuously sent.

[0111] The codec **200** may request a buffer of an eighth size (5K) corresponding to an eighth resolution (e.g., may be 5120*2160, hereinafter, referred to as 5K) greater than the seventh resolution (4K) from the codec controller **100** (S1312). For example, if it is possible to allocate a buffer with a size (e.g., b) greater than the resolution of the sent image frame, the codec may request a buffer with a size of b or more from the codec controller. For example, the codec may request the buffer of the eighth size (5K), which is greater than the buffer of the seventh size (4K), as fake resolution.

[0112] The codec controller **100** may allocate the buffer of the eighth size (5K) in the DPB **550** and

may send a seventh image frame FR7 with a seventh resolution (e.g., may be 3048*2160, hereinafter, referred to as 4K) to the codec **200** (S1313).

[0113] The codec **200** may send a seventh encoded image frame EN_FR7, which is obtained by encoding the seventh image frame FR7, to the codec controller **100** (S1314).

[0114] The codec controller **100** may allocate the buffer of the eighth size (5K) in the DPB **550** and may send the eighth image frame FR8 with the eighth resolution (5K) to the codec **200** (S1315).

[0115] The codec **200** may send an eighth encoded image frame EN_FR8, which is obtained by encoding the eighth image frame FR8, to the codec controller **100** (S1316).

[0116] As described in S1309 to S1316, by using the fake resolution, the number of interactions between the codec controller **100** and the codec **200** may be reduced while efficiently using the memory space. Through this, the memory space and performance may be balanced.

[0117] FIG. **14** is a flowchart for describing an operation of a video system according to exemplary embodiments of the present inventive concept. The specific values of the buffer size and frame resolution described in FIG. **14** are examples for convenience of explanation, and the exemplary embodiment of the present inventive concept is not limited thereto. A scenario in FIG. **14** corresponds to a further expanded exemplary embodiment of the scenario in FIG. **11**, and S1401 to S1410 in FIG. **14** correspond to steps corresponding to S1101 to S1110 in FIG. **11**, respectively. Therefore, in the following, the description will start from S1411.

[0118] Referring to FIGS. **7**, **11**, and **14**, an operation of a video system according to exemplary embodiments of the present inventive concept may be as follows.

[0119] The codec controller **100** may allocate the buffer of the sixth size (4K) in the DPB **550** and send a ninth image frame with a ninth resolution (e.g., may be 5120*2160, hereinafter, referred to as 5K) to the codec **200** (S1411). For example, an image frame with a resolution that is greater than the size of the allocated buffer may be sent.

[0120] The codec **200** may request a buffer of a tenth size (8K) corresponding to a tenth resolution (e.g., may be 64 k*64 k, hereinafter, referred to as 8K) that is greater than the ninth resolution (5K) from the codec controller **100** (S1412). For example, if it is possible to allocate a buffer with a size (e.g., b) greater than the resolution of the sent image frame, the codec may request a buffer with a size of b or more from the codec controller **100**. For example, the codec **200** may request the buffer of the tenth size (8K), which is greater than the buffer of the ninth size (5K), as fake resolution.

[0121] The codec controller **100** may allocate the buffer of the tenth size (8K) in the DPB **550** and may send the ninth image frame FR9 with the ninth resolution (5K) to the codec **200** (S1413).

[0122] The codec **200** may send a ninth encoded image frame EN_FR9, which is obtained by encoding the ninth image frame FR9, to the codec controller **100** (S1414).

[0123] The codec controller **100** may allocate the buffer of the tenth size (8K) in the DPB **550** and may send the tenth image frame FR10 with the tenth resolution (8K) to the codec **200** (S1415).

[0124] The codec **200** may send a tenth encoded image frame EN_FR10, which is obtained by encoding the tenth image frame FR10, to the codec controller **100** (S1416).

[0125] As described in S1409 to S1416, by using the fake resolution, the number of interactions between the codec controller **100** and the codec **200** may be reduced while efficiently using the memory space. Through this, the memory space and performance may be balanced.

[0126] FIG. **15** is a diagram for describing an electronic device including a video system according to exemplary embodiments of the present inventive concept.

[0127] Referring to FIG. **15**, an electronic device **1** including a video system according to exemplary embodiments of the present inventive concept may be a smartphone, but the present inventive concept is not limited thereto.

[0128] For example, the electronic device **1** including the video system according to exemplary embodiments of the present inventive concept may be implemented as a television (TV), a digital TV (DTV), an internet protocol TV (IPTV), a personal computer (PC), a desktop computer, a laptop computer, a computer workstation, a tablet PC, a video game platform (or a video game

console), a server, or a portable electronic device.

[0129] The portable electronic device may also be implemented as, for example, a smart phone, a personal digital assistant (PDA), an enterprise digital assistant (EDA), a digital still camera, a digital video camera, a portable multimedia player (PMP), a personal navigation device or portable navigation device (PND), a mobile internet device (MID), a wearable computer, an Internet of Things (IoT) device, an Internet of Everything (IoE) device, or an e-book.

[0130] FIG. 16 is a diagram describing a network system between electronic devices including a video system according to exemplary embodiments of the present inventive concept.

[0131] Referring to FIG. 16, a network system may include an image providing device 1 and an image receiving device 2.

[0132] The image providing device 1 may provide image signals including video and images to the image receiving device 2 through a network 3 in the streaming form, and the image receiving device 2 may receive and reproduce the image signals from the image providing device 1. The image providing device 1 may encode the image signal by removing and compressing redundant information in the image signal to a point where it might not be visually detected. The image providing device 1 may efficiently store, send, and manage the image signals by encoding the image signals within a predetermined compression rate range. The image signals may include a real-time encoded image including a screen mirroring image, a video conference image, and a game image.

[0133] The image providing device 1 may include various image source devices such as a TV, a PC, a smart phone, a tablet, a set-top box, a game console, and a server, and the image receiving device 2 may include various image playback devices such as a TV, a smart phone, a tablet, and a PC.

[0134] It is obvious to those skilled in the art that the image providing device 1 and the image receiving device 2 are not limited to a specific type of device. The image providing device 1 and the image receiving device 2 may transmit and receive the image signals through the network 3. According to various exemplary embodiments of the present inventive concept, the network 3 connecting the image providing device 1 and the image receiving device 2 to each other to facilitate the transmission of image signals may include, for example, a near-field communication network and a cellular network, such as Wireless Fidelity (WiFi), a next-generation telecommunications network, the Internet, or a telecommunications network such as a computer network (e.g., LAN or WAN), and may communicate based on the IP communication protocol. The cellular network may include, for example, Global System for Mobile Communications (GSM), Enhanced Data GSM Environment (EDGE), Code Division Multiple Access (CDMA), Time Division Multiplexing Access (TDMA), 5G, Long Term Evolution (LTE), and LTE-Advance (LTE-A). The network 3 may include connections of network elements such as hubs, bridges, routers, switches, and gateways. The network 3 may include one or more connected networks, including public networks such as the Internet and private networks such as corporate private networks, such as a multi-network environment. Access to the network 3 may be provided via one or more wired or wireless access networks. Furthermore, the network 3 may support an Internet of Things (IoT) network that exchanges and processes information between distributed components such as objects.

[0135] While the present inventive concept has been described with reference to embodiments thereof, it will be understood by those of ordinary skill in the art that various changes in form and details may be made thereto without departing from the spirit and scope of the present inventive concept.

Claims

1. A system on chip (SoC) comprising: a codec configured to encode a plurality of image frames; and a codec controller configured to send the plurality of image frames to the codec and to allocate

a buffer for encoding the plurality of image frames in a memory, wherein the codec controller sends a first image frame with a first resolution to the codec without allocating the buffer, the codec requests the codec controller to allocate a first buffer of a first size that corresponds to the first resolution, in response to receiving the first image frame, the codec controller allocates the first buffer to the memory and sends the first image frame to the codec, the codec encodes the first image frame by using the first buffer and then sends the encoded first image frame to the codec controller, the codec controller allocates the first buffer to the memory and sends a second image frame with a second resolution that is smaller than the first resolution to the codec, the codec requests the codec controller to allocate a second buffer of a second size that is smaller than the first size, in response to receiving the second image frame, the codec controller allocates the second buffer to the memory and sends the second image frame to the codec, and the codec encodes the second image frame by using the second buffer and then sends the encoded second image frame to the codec controller.

2. The system on chip (SoC) of claim 1, wherein the second size is a size corresponding to the second resolution.

3. The system on chip (SoC) of claim 2, wherein the codec controller allocates the second buffer to the memory and sends a third image frame with a third resolution that is greater than the second resolution to the codec, the codec requests the codec controller to allocate a third buffer of a third size corresponding to the third resolution, in response to receiving the third image frame, the codec controller allocates the third buffer to the memory and sends the third image frame to the codec, and the codec encodes the third image frame by using the third buffer and then sends the encoded third image frame to the codec controller.

4. The system on chip (SoC) of claim 1, wherein the second size is a size corresponding to a third resolution that is greater than the second resolution.

5. The system on chip (SoC) of claim 4, wherein the codec controller allocates the second buffer to the memory and sends a third image frame with the third resolution to the codec, and the codec encodes the third image frame by using the second buffer and then sends the encoded third image frame to the codec controller.

6. The system on chip (SoC) of claim 5, wherein the codec controller allocates the second buffer to the memory and sends a fourth image frame with the third resolution to the codec, the codec requests the codec controller to allocate a fourth buffer of a fourth size that corresponds to a fourth resolution that is greater than the third resolution, in response to receiving the fourth image frame, the codec controller allocates the fourth buffer to the memory and sends the fourth image frame to the codec, the codec encodes the fourth image frame by using the fourth buffer and then sends the encoded fourth image frame to the codec controller, the codec controller allocates the fourth buffer to the memory and sends a fifth image frame with the fourth resolution to the codec, and the codec encodes the fifth image frame by using the fourth buffer and then sends the encoded fifth image frame to the codec controller.

7. The system on chip (SoC) of claim 5, wherein the codec controller allocates the second buffer to the memory and sends a fourth image frame with a fourth resolution that is greater than the third resolution to the codec, the codec requests the codec controller to allocate the first buffer of the first size that corresponds to the first resolution that is greater than the fourth resolution, in response to receiving the fourth image frame, the codec controller allocates the first buffer to the memory and sends the fourth image frame to the codec, the codec encodes the fourth image frame by using the first buffer and then sends the encoded fourth image frame to the codec controller, the codec controller allocates the first buffer to the memory and sends a fifth image frame with the first resolution to the codec, and the codec encodes the fifth image frame by using the first buffer and then sends the encoded fifth image frame to the codec controller.

8. The system on chip (SoC) of claim 1, wherein the buffer is a decoded picture buffer (DPB).

9. The system on chip (SoC) of claim 1, wherein the codec sequentially decodes a plurality of

encoded image frames, the codec controller sends the plurality of encoded image frames to the codec and allocates a decoding buffer for decoding the plurality of encoded image frames in a memory, the codec controller sends a first encoded image frame with a third resolution to the codec without allocating the decoding buffer, the codec requests the codec controller to allocate a first decoding buffer that corresponds to the third resolution, in response to receiving the first encoded image frame, the codec controller allocates the first decoding buffer to the memory and sends the first encoded image frame to the codec, the codec decodes the first encoded image frame using the first decoding buffer and then sends the decoded first encoded image frame to the codec controller, the codec controller allocates the first decoding buffer to the memory and sends a second encoded image frame with a fourth resolution that is smaller than the third resolution to the codec, the codec requests the codec controller to allocate a second decoding buffer of a fourth size that is smaller than the third size, in response to receiving the second encoded image frame, the codec controller allocates the second decoding buffer to the memory and sends the second encoded image frame to the codec, and the codec decodes the second encoded image frame by using the second decoding buffer and then sends the decoded second encoded image frame to the codec controller.

10. A method for operating a system on chip (SoC) including a codec that is configured to sequentially encode a plurality of image frames, and a codec controller that is configured to send the plurality of image frames to the codec and allocate a buffer for encoding the plurality of image frames in a memory, the method comprising: sending a first image frame with a first resolution to the codec without allocating the buffer, through the codec controller; requesting, through the codec, the codec controller to allocate a first buffer of a first size corresponding to the first resolution, in response to receiving the first image frame; allocating the first buffer to the memory and sending the first image frame to the codec, through the codec controller; encoding the first image frame by using the first buffer and then sending the encoded first image frame to the codec controller, through the codec; allocating the first buffer to the memory and sending a second image frame with a second resolution that is smaller than the first resolution to the codec, through the codec controller; requesting the codec controller to allocate a second buffer of a second size that is smaller than the first size, in response to receiving the second image frame; allocating the second buffer to the memory and sending the second image frame to the codec; and encoding the second image frame by using the second buffer and then sending the encoded second image frame to the codec controller.

11. The method of claim 10, wherein the second size is a size that corresponds to the second resolution.

12. The method of claim 11, further comprising: allocating the second buffer to the memory and sending a third image frame with a third resolution that is greater than the second resolution to the codec, through the codec controller; requesting the codec controller to allocate a third buffer of a third size that corresponds to the third resolution, in response to receiving the third image frame, through the codec; allocating the third buffer to the memory and sending the third image frame to the codec, through the codec controller; and encoding the third image frame by using the third buffer and then sending the encoded third image frame to the codec controller, through the codec.

13. The method of claim 10, wherein the second size is a size that corresponds to a third resolution that is greater than the second resolution.

14. The method of claim 13, further comprising: allocating the second buffer to the memory and sending a third image frame with the third resolution to the codec, through the codec controller; and encoding the third image frame by using the second buffer and then sending the encoded third image frame to the codec controller, through the codec.

15. The method of claim 14, further comprising: allocating the second buffer to the memory and sending a fourth image frame with the third resolution to the codec, through the codec controller; requesting the codec controller to allocate a fourth buffer of a fourth size that corresponds to a fourth resolution that is greater than the third resolution, in response to receiving the fourth image

frame, through the codec; allocating the fourth buffer to the memory and sending the fourth image frame to the codec, through the codec controller; encoding the fourth image frame using the fourth buffer and then sending the encoded fourth image frame to the codec controller, through the codec; allocating the fourth buffer to the memory and sending a fifth image frame with the fourth resolution to the codec, through the codec controller; and encoding the fifth image frame by using the fourth buffer and then sending the encoded fifth image frame to the codec controller, through the codec.

16. The method of claim 14, further comprising: allocating the second buffer to the memory and sending a fourth image frame with a fourth resolution that is greater than the third resolution to the codec, through the codec controller; requesting the codec controller to allocate the first buffer of the first size that corresponds to the first resolution that is greater than the fourth resolution, in response to receiving the fourth image frame, through the codec; allocating the first buffer to the memory and sending the fourth image frame to the codec, through the codec controller; encoding the fourth image frame using the first buffer and then sending the encoded fourth image frame to the codec controller, through the codec; allocating the first buffer to the memory and sending a fifth image frame with the first resolution to the codec, through the codec controller; and encoding the fifth image frame by using the first buffer and then sending the encoded fifth image frame to the codec controller, through the codec.

17. The method of claim 10, wherein the buffer is a decoded picture buffer (DPB).

18. A video system comprising: a codec configured to encode a plurality of image frames; a codec controller configured to send the plurality of image frames to the codec and to allocate a buffer for encoding the plurality of image frames in a memory; a camera configured to send the plurality of image frames to the codec controller; and a display configured to display the plurality of image frames, wherein the codec controller sends a first image frame with a first resolution to the codec without allocating the buffer, the codec requests the codec controller to allocate a first buffer of a first size that corresponds to the first resolution, in response to receiving the first image frame, the codec controller allocates the first buffer to the memory and sends the first image frame to the codec, the codec encodes the first image frame by using the first buffer and then sends the encoded first image frame to the codec controller, the codec controller allocates the first buffer to the memory and sends a second image frame with a second resolution that is smaller than the first resolution to the codec, the codec requests the codec controller to allocate a second buffer of a second size that is smaller than the first size, in response to receiving the second image frame, the codec controller allocates the second buffer to the memory and sends the second image frame to the codec, and the codec encodes the second image frame by using the second buffer and then sends the encoded second image frame to the codec controller.

19. The video system of claim 18, wherein the second size is a size that corresponds to the second resolution, the codec controller allocates the second buffer to the memory and sends a third image frame with a third resolution that is greater than the second resolution to the codec, the codec requests the codec controller to allocate a third buffer of a third size that corresponds to the third resolution, in response to receiving the third image frame, the codec controller allocates the third buffer to the memory and sends the third image frame to the codec, and the codec encodes the third image frame by using the third buffer and then sends the encoded third image frame to the codec controller.

20. The video system of claim 18, wherein the second size is a size that corresponds to a third resolution that is greater than the second resolution, the codec controller allocates the second buffer to the memory and sends a third image frame with the third resolution to the codec, and the codec encodes the third image frame by using the second buffer and then sends the encoded third image frame to the codec controller.
